American Computer Science League

2021-2022 ● Contest 2: Fibonacci Cypher ● Intermediate Division

PROBLEM: Use this Fibonacci cypher to encode a given message.

The Fibonacci sequence can start with any two non-negative integers and every term after that is the sum of the two previous numbers. Therefore, if the first two numbers are 3 and 7, the sequence is 3, 7, 10, 17, 27, ... A Fibonacci cypher uses these numbers and an inputted 'key' (a lower-case letter in the alphabet) to encode a message into a series of numbers. Another person can decipher the message by knowing the 'key' that was used and the starting 2 numbers of the sequence. For each character in the message, integers are generated by using the following steps:

1. Find the Fibonacci number with the same index as the character (e.g. using the sequence above, the 1st character uses the number 3 and the 5th character uses the number 27).
2. By using the inputted key, find a new letter that is offset by the Fibonacci number found in Step 1. Wrap around to the beginning of the alphabet if necessary (e.g. if the key = 'p' and the Fibonacci number is 17, then the new letter is 'g').
3. Find the number for each letter in the encoded message by adding the ASCII code of the character in the string to 3 times the ASCII code of the letter found in Step 2.

In order for the Fibonacci numbers to not become too large, every message will be encoded in blocks of no more than 20 characters so the 21st character would use the first Fibonacci number again as would the 41st. We guarantee the message will not be longer than 100 characters.

EXAMPLE: For the string "ACSL c2", the initial Fibonacci numbers 3 and 7, and the letter 'h' as the 'key', the numbers are found as shown:

Fibonacci offset	ASCII code	Calculation
h + 3 = k = 107	A = 65	65 + 3 * 107 = 386
h + 7 = o =111	C = 67	67 + 3 * 111 = 400
h + 10 = r = 114	S = 83	83 + 3 * 114 = 425
h + 17 = y = 121	L = 76	76 + 3 * 121 = 439
h + 27 = i = 105	space = 32	32 + 3 * 105 = 347
h + 44 = z = 122	c = 99	99 + 3 * 122 = 465
h + 71 = a = 97	2 = 50	50 + 3 * 97 = 341

The encoded message is "386 400 425 439 347 465 341".

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INPUT: There are 5 sets of data. Each set has four pieces of information. 1&2) the two starting non-negative integers of the Fibonacci sequence, 3) the 'key' which is a single lower-case alphabetic character, and 4) a string representing the actual message to be encoded. Any typeable character on the keyboard except for double quotes may be included in the message.

OUTPUT: For each line of data, print the integers separated by a single space that represent the encoded message.

SAMPLE INPUT:

```
3 7 h ACSL c2
4 9 s Python Programming is easier than programming in Java.
2 5 a Fibonacci Numbers are found in many places including the
Mandelbrot Set.
0 1 j Help ME figure out how to solve this problem!
4 8 z It is 9:30 in the morning EST, but 6:30 on the West Coast.
```

SAMPLE OUTPUT:

```
1. 386 400 425 439 347 465 341
2. 437 415 422 437 405 470 341 404 480 456 469 480 406 439 403
   462 416 421 389 435 472 326 407 430 409 465 410 438 398 461
   470 463 419 362 406 471 417 421 471 427 466 403 411 443 397
   392 414 434 398 419 463 484 406 376
3. 367 411 410 438 458 403 462 399 399 335 384 435 442 458 425
   429 463 326 448 468 398 338 414 438 465 416 463 332 399 413
   338 427 430 470 445 347 460 402 448 453 398 421 344 432 458
   405 471 417 394 408 416 421 365 476 428 416 380 371 448 464
   397 407 420 425 462 417 479 332 377 404 422 364
4. 390 422 429 436 359 410 411 389 405 447 430 468 474 416 389
   465 432 467 380 407 429 440 353 440 438 365 457 468 411 460
   428 383 476 419 462 469 347 463 462 414 416 429 422 433 360
5. 373 428 356 453 421 398 363 364 375 390 332 459 476 386 470
   446 431 338 457 477 414 422 429 458 409 398 375 389 408 386
   332 452 483 470 386 396 388 357 396 398 411 422 356 464 410
   467 338 393 425 457 416 386 433 465 451 457 446 352
```

TEST DATA

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TEST INPUT:

```
1 2 a Fibonacci cyphers are fun to encode and decode.
4 7 x You may only use C++, Java, and Python in HackerRank.
7 10 p Madam, I'm Adam is one of the most common palindromes.
16 9 l What happens when the first number is greater than the
second?
11 13 q The ratio of 2 adjacent Fibonacci numbers in the
sequence approaches the Golden number, phi, which is
approx. 1.618...
```

TEST OUTPUT:

1. 364 402 398 417 425 427 453 414 405 356 432 487 442 431 467
   438 436 386 403 405 395 329 402 423 425 362 470 426 332 425
   443 465 441 427 467 356 418 464 406 323 394 398 399 417 415
   431 400
2. 383 414 432 368 400 442 475 371 444 422 471 436 350 468 424
   401 359 412 355 340 338 335 389 433 409 442 398 371 430 422
   463 347 398 472 425 404 438 455 344 402 404 335 387 433 390
   452 455 453 415 409 473 422 364
3. 434 463 409 436 421 359 323 421 342 424 392 404 463 463 424
   377 429 448 353 429 467 467 341 450 414 347 407 452 404 347
   469 450 478 482 347 444 435 442 430 429 467 398 421 436 420
   420 401 448 417 426 469 440 478 412
4. 381 455 418 464 377 395 409 469 457 401 431 412 326 464 419
   437 437 371 458 461 395 383 423 453 459 406 428 389 455 417
   430 395 395 459 347 441 442 371 445 471 395 448 437 449 459
   323 428 461 442 410 353 413 398 446 347 451 428 438 453 467
   394 414
5. 378 404 434 326 480 418 464 435 450 362 441 423 344 344 377
   397 406 451 409 417 395 410 449 326 436 426 446 441 449 427
   429 420 417 326 455 417 415 443 413 432 409 332 438 404 398
   437 452 431 371 445 431 434 429 395 455 399 407 377 409 430
   406 414 444 391 465 425 449 445 371 446 434 422 344 365 456
   408 406 446 422 350 404 417 442 392 467 435 392 362 451 434
   435 365 344 413 449 405 405 449 344 423 409 332 430 406 478
   435 459 450 385 362 379 367 366 343 401 346 352 391

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题目：使用斐波那契加密方法给以下特定的信息编码加密。

斐波那契数列的前两项可以**是任意**的两个**非负整数，**且后面**每一项都是前两项**之**和。即如果数列的**最前面的两项**为** 3 和 7，**则该数列是** 3, 7, 10, 17, 27, ... **斐波那契**加密方法**使用这些数字和一个输入“ 密钥 ”(字母表中的一个小写字母)将一**条**信息编码成一**串**数字。**其他**人**可通过已使用的**“ 密钥 ”和数列的前两项，**来**破译这**条**信息。对于信息中的每个**字符，都可以**根据以下步骤生成对应的整数：**

1. 找出与该字符具有相同索引的斐波那契数（如使用上述数列，第一个字符用数字 3 来表示，第五个字符用数字 27 来表示。）
2. 根据输入的密钥，找出对步骤 1 中斐波那契数进行偏移操作的新字母。如有必要，可以重新绕到字母表的开头处（如假设密钥是“p”，斐波那契数是 17，那么新的字母即为 ‘g’）。
3. 通过将字符串中该字符的 ASCII 码与步骤 2 中该字母的 ASCII 码的 3 倍相加，即可以找出编码信息中每个字符的加密数字。

为了保证斐波那契数不会变得太大，每条信息将被编码为不超过 20 个字符的代码块，因此第 21 个字符可以再次使用第一个斐波那契数，第 41 个字符同样如此。我们保证每条信息的长度不超过 100 个字符。

示例：对于字符串 “ACSL c2”，斐波那契数列的前两项为 3 和 7，字母 ‘h’ 为 “ 密钥 ”，编码数字可以通过以下方式获得：

斐波那契偏移	ASCII 码	计算结果
h + 3 = k = 107	A = 65	65 + 3 * 107 = 386
h + 7 = o =111	C = 67	67 + 3 * 111 = 400
h + 10 = r = 114	S = 83	83 + 3 * 114 = 425
h + 17 = y = 121	L = 76	76 + 3 * 121 = 439
h + 27 = i = 105	space = 32	32 + 3 * 105 = 347
h + 44 = z = 122	c = 99	99 + 3 * 122 = 465
h + 71 = a = 97	2 = 50	50 + 3 * 97 = 341

编码后的信息为 “386 400 425 439 347 465 341”。

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输入: 共有 5 组数据，每组包**含** 4 部分信息：1&2) 斐波那契数列中起始的两个非负整数，3) “密钥”，即一个小写字母，4) 一条需要编码的字符串信息。除双引号外，键盘上的任意字符都可能在信息中出现。

输出:每行数据都输出整数，每个整数之间由空格隔开，表示编码后的信息。

样本输入:

```
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4 9 s Python Programming is easier than programming in Java.
2 5 a Fibonacci Numbers are found in many places including the
Mandelbrot Set.
0 1 j Help ME figure out how to solve this problem!
4 8 z It is 9:30 in the morning EST, but 6:30 on the West Coast.
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预期输出:

1. 386 400 425 439 347 465 341
2. 437 415 422 437 405 470 341 404 480 456 469 480 406 439 403
   462 416 421 389 435 472 326 407 430 409 465 410 438 398 461
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   392 414 434 398 419 463 484 406 376
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   429 463 326 448 468 398 338 414 438 465 416 463 332 399 413
   338 427 430 470 445 347 460 402 448 453 398 421 344 432 458
   405 471 417 394 408 416 421 365 476 428 416 380 371 448 464
   397 407 420 425 462 417 479 332 377 404 422 364
4. 390 422 429 436 359 410 411 389 405 447 430 468 474 416 389
   465 432 467 380 407 429 440 353 440 438 365 457 468 411 460
   428 383 476 419 462 469 347 463 462 414 416 429 422 433 360
5. 373 428 356 453 421 398 363 364 375 390 332 459 476 386 470
   446 431 338 457 477 414 422 429 458 409 398 375 389 408 386
   332 452 483 470 386 396 388 357 396 398 411 422 356 464 410
   467 338 393 425 457 416 386 433 465 451 457 446 352

测试数据

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测试输入:

```
1 2 a Fibonacci cyphers are fun to encode and decode.
4 7 x You may only use C++, Java, and Python in HackerRank.
7 10 p Madam, I'm Adam is one of the most common palindromes.
16 9 l What happens when the first number is greater than the
second?
11 13 q The ratio of 2 adjacent Fibonacci numbers in the
sequence approaches the Golden number, phi, which is
approx. 1.618...
```

测试输出:

1. 364 402 398 417 425 427 453 414 405 356 432 487 442 431 467
   438 436 386 403 405 395 329 402 423 425 362 470 426 332 425
   443 465 441 427 467 356 418 464 406 323 394 398 399 417 415
   431 400
2. 383 414 432 368 400 442 475 371 444 422 471 436 350 468 424
   401 359 412 355 340 338 335 389 433 409 442 398 371 430 422
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   469 450 478 482 347 444 435 442 430 429 467 398 421 436 420
   420 401 448 417 426 469 440 478 412
4. 381 455 418 464 377 395 409 469 457 401 431 412 326 464 419
   437 437 371 458 461 395 383 423 453 459 406 428 389 455 417
   430 395 395 459 347 441 442 371 445 471 395 448 437 449 459
   323 428 461 442 410 353 413 398 446 347 451 428 438 453 467
   394 414
5. 378 404 434 326 480 418 464 435 450 362 441 423 344 344 377
   397 406 451 409 417 395 410 449 326 436 426 446 441 449 427
   429 420 417 326 455 417 415 443 413 432 409 332 438 404 398
   437 452 431 371 445 431 434 429 395 455 399 407 377 409 430
   406 414 444 391 465 425 449 445 371 446 434 422 344 365 456
   408 406 446 422 350 404 417 442 392 467 435 392 362 451 434
   435 365 344 413 449 405 405 449 344 423 409 332 430 406 478
   435 459 450 385 362 379 367 366 343 401 346 352 391